

RODRIGO GOMES DE ARAÚJO

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SUMMARY

MSc Computer Science student (graduating **July 2027**) focused on systems where **correctness and robustness** are key. Built a **proof-of-work blockchain** in Rust and is currently extending session types to **enforce correct usage of cryptographic resources** at compile-time. Works well **autonomously** and **asynchronously** and emphasizes **testing, documentation and API** in the development process. Comfortable working remotely and async. Based in Porto, Portugal.

EDUCATION

Master's <i>Computer Science</i>	Sep 2025 – Present (Jul 2027) <i>University of Porto (FCUP)</i>
Bachelor's <i>Computer Science and Engineering</i>	Sep 2022 – Jul 2025 <i>University of Porto (FEUP)</i>

RESEARCH

CaST (Cryptographically-aware Session Types) <i>Master's Thesis</i>	Jul 2026 – Present (Jul 2027) <i>FreeST, Haskell, Type Systems, Applied Cryptography</i>
- Extending the expressiveness of session types to capture cryptographic resource usage, so that its misuse in protocols is detected and rejected at compile-time. Resource tracking composes across nested subprotocols of arbitrary depth. - Formalizing the extension and providing a soundness proof, integrating it into the FreeST compiler and applying it to smart contracts and protocol specification.	

PROJECTS

PoW Blockchain for Decentralized Auctions | Rust, Tokio, LibP2P, Dalek, Docker

- Extensive **research on the state of the art** of Bitcoin, Ethereum and Solana and investigation regarding the security and resilience to failure of these systems.
- Developed a DHT overlay based on Kademlia for **decentralized network construction**.
- PoW Blockchain with **Blake2b hashing, Ed25519 signatures** and **Merkle Trees** for transaction validation.
- **Client for auctions** using the blockchain layer as a public ledger.
- Docker **adversarial network simulation** with clients and bootstrap, well behaved and malicious nodes.
- Developed as Data Systems Security final project and graded 19.1/20.

Verified-by-Construction Development Workflow | Rust, Gitoxide, Ed25519, IPLD/DAG-CBOR, Serde

- Development lifecycle that **enforces verification by construction**: ticket → **frozen specification** → implementation → **immutable submitted version** → verification → review → accepted.
 - Specifications are **frozen before implementation starts**, so every change is verified against exactly what was agreed and **nothing reaches the main branch without passing the verification pipeline**. If the main branch moved in the meantime, the contribution becomes **Conflicted** and is sent back to be resolved, **re-verified and re-reviewed**, so the guarantees hold even after merges.
- Lifecycle states (Version<Implemented>, Version<Verified>, Version<Conflicted>) modelled as **types in Rust**, making invalid transitions unrepresentable; verified records and their evidence **cannot be silently replaced or reassigned**.
- Verification **bound to the exact commit it evaluates** via thread-safe **Gitoxide** clones with **caching**, enabling concurrent, asynchronous verification without ever checking the wrong version.
- **Git-backed persistence layer** storing versioned DAG-CBOR/IPLD records directly in Git trees, giving **immutable, content-addressed** provenance from spec to accepted code.
- **Ed25519-based authorization** with tiered authority (agents, operators, owners), with signed requests bound to project, action, subject and expected repository state.

AWARDS

2nd Place in IEEE RetroJam 2025 IEEE UP Student Branch	Oct 2025
Game from scratch in 48 hours (Rust and Raylib), learning how to reduce a problem to its core requirements.	
Academic Merit Awards	2020, 2021, 2022
Participation in RoboCup 2016 RoboCup Federation	Jun 2016
1st Place in the National Robotics Championship Instituto Politécnico de Bragança	May 2016

TECHNICAL SKILLS

Programming Languages: Rust, Haskell, C/C++, OCaml, Java, SQL
Frameworks & Tools: Git, UNIX, Tokio, Docker, Plotly, POSIX, STM, gRPC, LibP2P, Dalek, PostgreSQL, Terraform, Raylib and OpenMP
Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)